

Table 2. Summary of truthfulness evaluations (DistilGPT-2, Qwen-2.5-1.5B/7B/14B)

| Model         | Method     | T.I (%) (up)           | True (%) (up)   | Info (%) (up)    | MLU (%) (up)   |
|---------------|------------|------------------------|-----------------|------------------|----------------|
| GPT-2-0.5B    | Original   | 14.06 +/- 7.59         | 18.75 +/- 9.77  | 5.00 +/- 10.80   | 0.00 +/- 0.00  |
|               | A-LQR      | 18.75 +/- 9.77         | 18.75 +/- 9.77  | 100.00 +/- 0.00  | 0.00 +/- 0.00  |
| 2.5-1.5B-ours | S-PID      | <b>31.25 +/- 11.31</b> | 31.25 +/- 11.50 | 100.00 +/- 0.00  | 4.17 +/- 4.08  |
|               | H-infinity | 18.75 +/- 9.77         | 18.75 +/- 9.77  | 100.00 +/- 0.00  | 0.00 +/- 0.00  |
|               | Original   | 25.00 +/- 10.82        | 25.00 +/- 10.80 | 100.00 +/- 0.00  | 8.33 +/- 5.64  |
| 2.5-7B-ours   | A-LQR      | 0.00 +/- 0.00          | 0.00 +/- 0.00   | 0.00 +/- 0.00    | 20.83 +/- 8.29 |
|               | S-PID      | <b>31.25 +/- 11.31</b> | 31.25 +/- 11.50 | 100.00 +/- 0.00  | 8.33 +/- 5.64  |
|               | H-infinity | 0.00 +/- 0.00          | 0.00 +/- 0.00   | 6.25 +/- 6.05    | 20.83 +/- 8.29 |
| 2.5-14B-ours  | Original   | 33.33 +/- 19.23        | 33.33 +/- 19.20 | 100.00 +/- 0.00  | 7.50 +/- 17.12 |
|               | A-LQR      | 0.00 +/- 0.00          | 0.00 +/- 0.00   | 100.00 +/- 0.00  | 0.00 +/- 0.00  |
|               | S-PID      | <b>16.67 +/- 15.12</b> | 16.67 +/- 15.20 | 100.00 +/- 0.00  | 5.00 +/- 15.31 |
| 2.5-14B       | H-infinity | -- +/- --              | -- +/- --       | -- +/- --        | -- +/- --      |
|               | Original   | 12.50 +/- 11.66        | 12.50 +/- 11.60 | 100.00 +/- 0.00  | 0.00 +/- 0.00  |
|               | A-LQR      | <b>0.00 +/- 0.00</b>   | 0.00 +/- 0.00   | 100.00 +/- 0.00  | 0.00 +/- 0.00  |
|               | S-PID      | <b>0.00 +/- 0.00</b>   | 0.00 +/- 0.00   | 100.00 +/- 20.40 | 0.00 +/- 0.00  |
|               | H-infinity | -- +/- --              | -- +/- --       | -- +/- --        | -- +/- --      |